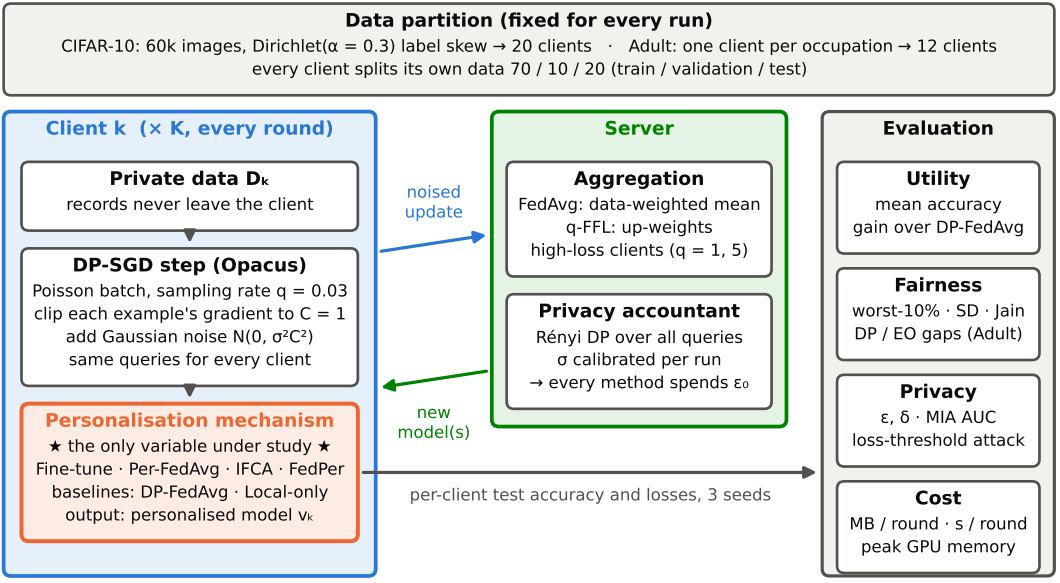
Data partition (fixed for every run)
CIFAR-10: 60k images, Dirichlet(α = 0.3) label skew → 20 clients · Adult: one client per occupation → 12 clients
every client splits its own data 70 / 10 / 20 (train / validation / test)
Client k (× K, every round)
Private data Dₖ
records never leave the client
DP-SGD step (Opacus)
Poisson batch, sampling rate q = 0.03
clip each example's gradient to C = 1
add Gaussian noise N(0, σ²C²)
same queries for every client
Personalisation mechanism
★ the only variable under study ★
Fine-tune · Per-FedAvg · IFCA · FedPer
baselines: DP-FedAvg · Local-only
output: personalised model vₖ
noised update
new model(s)
Server
Aggregation
FedAvg: data-weighted mean
q-FFL: up-weights high-loss clients (q = 1, 5)
Privacy accountant
Rényi DP over all queries
σ calibrated per run
→ every method spends ε₀
per-client test accuracy and losses, 3 seeds
Evaluation
Utility
mean accuracy gain over DP-FedAvg
Fairness
worst-10% · SD · Jain
DP / EO gaps (Adult)
Privacy
ε, δ · MIA AUC
loss-threshold attack
Cost
MB / round · s / round
peak GPU memory